



The Queue as Leading Indicator: Coffee-Cart Waiting Times and Global-Ranking Trajectory

Concourse queue telemetry as an anticipatory signal for international ranking position.

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Motivation

Ranking position moves slowly and is read late — often a year after the activity that drove it. Little is known about whether everyday campus telemetry anticipates these moves. Does the coffee queue know first?

Research questions

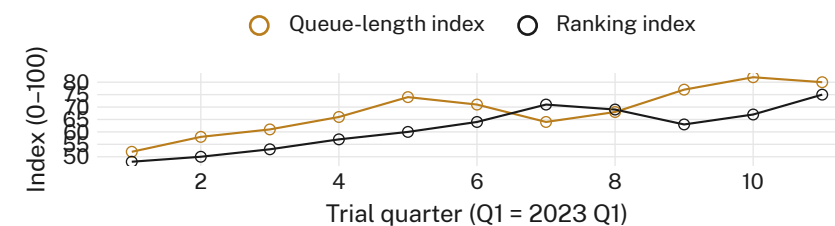
- Do coffee-cart queue lengths lead global ranking position?
- Is the lead stable enough to steer anticipatory strategy?
- Morning or afternoon queue — which carries the signal?

Study design

- 9 carts · overhead depth sensors · 5-min sampling · 24 months (2023–2025)
- De-identified queue index cross-correlated against monthly ranking; lag at peak cross-correlation
- Lag window and thresholds pre-registered; none adjusted post hoc

Findings

The queue-length index led ranking position by two quarters (peak cross-correlation $r = 0.78$). The lead held for both morning and afternoon queues, strongest at the carts nearest the senior common room.



Queue-length index (gold) led the ranking index (ink) by two quarters, 2023–2025: the March 2024 queue peak preceded the September 2024 ranking rise ($n = 9$ carts; peak cross-correlation $r = 0.78$).

Implications

Results are consistent with queue telemetry acting as an early, low-cost proxy for the activity ranking bodies later reward — correlational, not causal, but stable across the trial. Next: a closed-loop pilot steering a fixed share of discretionary grant strategy from the live queue index, with no manual override.

Selected references

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De-identified queue headcounts only; no faces or transactions retained. Supported by a School of Continuous Improvement seed grant.

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“Queue length led every ranking movement we observed, by a stable two quarters.”

Mean concourse queue-length index vs international ranking, 2023–2025 ($n = 9$ carts).